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Activity 1

Linear search (worst case analysis)

Worst case :- The element is not present or when the ~~ee~~ number of comparsion is more

Number of comparsion: n
Time complexity: O(n)

code (c program)

```
# include <stdio.h>
int main(){
    int a[100], n, key, i;
    printf("Enter the number of elements:");
    scanf(%d, & a[i]); }
    for (i=0; i<n; i++){
        scanf (%d, & a[i]); }
    printf("Enter the element to search:");
    scanf("%d", & Key);
    for (i=0; i<n; i++){
      if (a[i] == key){
          print (" Element found at position
              %d\n", i+1); return 0; }}
      print (" Element not found \n");
      return 0; }
```

Input
Enter the number of elements: 5
Enter the elements: 10 12 13 60 78
Enter the element to search: 78

Output :- Element found at position: 4